

Experiment 08: Input & Output Manual

Integration of Third-Party Services in Web Application (Email, SMS, Payment)

Student & Laboratory Credentials

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- **Class / Division:** B.Tech Computer Engineering (Sem VI)
- **Academic Year:** 2026 – 2027
- **Course:** Skill Based Lab - Advanced Web Technology (SBL-AWT)
- **Institution:** Ramrao Adik Institute of Technology (RAIT), D Y Patil Deemed to be University, Nerul, Navi Mumbai
- **Problem Statement Domain:** HealthPulse — Digital Healthcare & Clinical Management Portal
- **Experiment Title:** Integration of Third-Party Services in Web Application (SMS Gateway, Payment Gateway, Email)

1. Experiment Aim

To design, implement, and integrate essential third-party cloud microservices into a healthcare web portal: 1. **Email Gateway (Nodemailer):** Automated medical prescription dispatch with live cloud SMTP preview generation. 2. **SMS Gateway (Twilio Protocol):** Two-factor authentication (2FA) one-time password (OTP) and clinical alert dispatch. 3. **Payment Gateway (Razorpay/Stripe Protocol):** End-to-end checkout order creation, transaction verification, and receipt capturing.

2. Input Specification

2.1 File System Input

File Name	Format	Role & Implementation Responsibility
server.js	Express.js / Node.js	Microservices hub configuring Nodemailer Ethereal SMTP transport, Twilio SMS protocol handler, Razorpay/Stripe checkout API, and real-time transaction event logger.
package.json	JSON	Dependencies declaration (<code>express</code> , <code>nodemailer</code> , <code>cors</code>) and execution scripts.
public/index.html	HTML5 / CSS3 / JS	Unified laboratory testing dashboard with trigger panels for Email, SMS, and Payment, connected to an active JSON transaction logs terminal.

2.2 Execution Command Input

```
# Navigate to the experiment directory
cd /home/ubuntu/SBL-EXPT/Asmit_25CE1051/Expt-08-ThirdParty-Services

# Step 1: Install dependencies
npm install

# Step 2: Start microservices server (Default Port: 8000)
node server.js

# Or with custom port:
PORT=8000 node server.js
```

2.3 Microservices API Input Schema

SERVICE / API ROUTE	HTTP	INPUT PARAMETERS (JSON PAYLOAD)	OUTPUT ACTION
/api/service/email	POST	{ patientEmail, patientName, doctorName, prescriptionNotes }	Sends SMTP e-mail & returns preview URL
/api/service/sms	POST	{ phone, patientName, appointmentTime }	Generates 6-digit OTP & Twilio Message SID
/api/service/payment/order	POST	{ amount, patientName, consultationType }	Creates Razorpay order in paisa with Order ID
/api/service/payment/verify	POST	{ orderId, paymentId, amount, patient }	Verifies signature & issues digital receipt

2.4 Sample Input Test Payloads

Test Vector 1: Email Prescription Dispatch

```
curl -X POST http://localhost:8000/api/service/email \
-H "Content-Type: application/json" \
-d '{
  "patientEmail": "asmit.jogdand@healthpulse.org",
  "patientName": "Asmit Jogdand",
  "doctorName": "Dr. Amarsinh V. Vidhate",
  "appointmentDate": "2026-10-05",
  "prescriptionNotes": "Tab Paracetamol 500mg TDS x 3 days, Tab Cetirizine 10mg OD x 5 days"
}'
```

Test Vector 2: SMS OTP Verification Alert

```
curl -X POST http://localhost:8000/api/service/sms \
-H "Content-Type: application/json" \
-d '{
  "phone": "9876543210",
  "patientName": "Asmit Jogdand",
  "appointmentTime": "10:30 AM"
}'
```

Test Vector 3: Payment Order Generation

```
curl -X POST http://localhost:8000/api/service/payment/order \
-H "Content-Type: application/json" \
-d '{
  "amount": 800,
  "patientName": "Asmit Jogdand",
  "consultationType": "Cardiology Super-Specialty OPD"
}'
```

3. Output Specification

3.1 Terminal Execution & Gateway Startup Logs

```
$ PORT=8000 node server.js
[HealthPulse] Third-Party Services Gateway running at http://localhost:8000
[HealthPulse] Student: Asmit Jogdand (25CE1051)
[HealthPulse Email] Ethereal SMTP Initialized. Test User: vmvnfbuhd4tlcgth@ethereal.email
[HealthPulse Gateway] Nodemailer Ethereal Cloud connection verified.
```

3.2 Microservice 1 Output: Email Dispatch Response

```
HTTP/1.1 200 OK
Content-Type: application/json; charset=utf-8

{
  "success": true,
  "message": "Clinical email successfully dispatched via SMTP gateway!",
  "data": {
    "service": "Email Gateway (Nodemailer)",
    "recipient": "asmit.jogdand@healthpulse.org",
    "messageId": "<5f7cb53a-26e4-9c4a-986f-0a540585d997@healthpulse.com>",
    "previewUrl": "https://ethereal.email/message/ar4ogQLEqXPhQbfqar4ohcAYPlgYhv4sAAAAAe2NjK1QAU0xuebsZSfheeA",
    "status": "Delivered",
    "timestamp": "9:31:49 AM"
  }
}
```

(Opening the generated [previewUrl](#) in any browser displays the full medical HTML prescription with clinical header, patient demographics, and doctor notes).

3.3 Microservice 2 Output: SMS Gateway Alert Response

```
HTTP/1.1 200 OK
Content-Type: application/json; charset=utf-8

{
  "success": true,
  "message": "SMS alert dispatched to 9876543210 successfully!",
  "data": {
    "service": "SMS Gateway (Twilio Integration Protocol)",
    "recipient": "9876543210",
    "sid": "SMrrismdhypn7k1267pp923",
    "otp": 334340,
    "body": "[HealthPulse] Hello Asmit Jogdand, your OTP for consultation verification is 334340. Appointment confirmed for 10:30 AM. Do not share this OTP.",
    "gatewayResponse": "200 OK - Delivered via Telecom Provider",
    "status": "Sent / Delivered",
    "timestamp": "9:32:06 AM"
  }
}
```

3.4 Microservice 3 Output: Payment Gateway Order & Capture Responses

A. Order Creation Response:

```
HTTP/1.1 200 OK
Content-Type: application/json; charset=utf-8

{
  "success": true,
  "message": "Payment order generated on payment gateway",
  "order": {
    "orderId": "order_hp_gdzhwhyxeh",
    "receiptNo": "rcpt_444012",
    "amount": 800,
    "currency": "INR",
    "amountInPaise": 80000,
    "patientName": "Asmit Jogdand",
    "consultationType": "Cardiology Super-Specialty OPD",
    "status": "Created",
    "keyId": "rzp_test_HPHealthPulse2026"
  }
}
```

B. Verification & Capture Response ([POST /api/service/payment/verify](#)):

```

HTTP/1.1 200 OK
Content-Type: application/json; charset=utf-8

{
  "success": true,
  "message": "Payment verified and captured successfully!",
  "receipt": {
    "service": "Payment Gateway (Razorpay/Stripe Protocol)",
    "orderId": "order_hp_gdzhwhyxeh",
    "paymentId": "pay_test_99214488",
    "amount": "₹800",
    "patient": "Asmit Jogdand",
    "verificationStatus": "SIGNATURE_VERIFIED_SUCCESSFUL",
    "status": "Payment Captured (PAID)",
    "timestamp": "9:32:06 AM"
  }
}

```

3.5 Unified Testing Dashboard Output Representation ([GET /](#))

```

+-----+
| [HealthPulse Portal]  SBL - Advanced Web Technology (Sem VI)          Student: Asmit (25CE1051) |
+-----+
|                                     |
|               ▢ THIRD-PARTY MICROSERVICES TESTING DASHBOARD          |
|           Integration Hub for Nodemailer (Email), Twilio (SMS), & Razorpay |
|                                     |
| +-- 1. Nodemailer Email Gateway --+ +-- 2. Twilio SMS OTP Gateway --+ +-- 3. Razorpay Checkout --+ |
| | Patient: [ Asmit Jogdand      ] | | Phone: [ +91 9876543210      ] | | Consultation Fee: ₹800 | |
| | Email:   [ asmit@hp.org       ] | | Time:   [ 10:30 AM           ] | | Patient: Asmit Jogdand | |
| | Notes:   [ Tab Paracetamol TDS ] | | Purpose: 2FA Verification OTP | | Mode: Super-Specialty | |
| | [ ✕ Dispatch Prescription ]    | | [ ▢ Dispatch SMS & OTP ]      | | [ 📄 Execute Checkout ] | |
| +-----+ +-----+ +-----+ |
|                                     |
| +-- Real-Time Gateway Transaction Logs Terminal -----+ |
| | [09:31:49 AM] [EMAIL] Status: Delivered | MessageID: <5f7cb53a...> | Preview: ethereal.email/.. |
| | [09:32:06 AM] [SMS]   Status: Delivered | SID: SMrrismdh... | Generated OTP: 334340 |
| | [09:32:06 AM] [PAY]   Status: Captured  | Order: order_hp_gdzh... | Amount: ₹800.00 PAID |
| +-----+ |
+-----+

```

3.6 Test Case Execution Results

Test Scenario	Microservice Triggered	Backend Validation & Gateway Logic	Observed Output Result	Status
Email Missing Recipient	POST /api/service/email (no email)	Rejects payload missing mandatory email	HTTP 400: "Recipient email is required"	PASS
Valid Email Dispatch	Complete email payload	Connects to SMTP; returns messageid	HTTP 200; Live Ethereal preview link returned	PASS
SMS Missing Phone	POST /api/service/sms (no phone)	Rejects payload missing cellular number	HTTP 400: "Phone number is required"	PASS
Valid SMS OTP Alert	Injects phone 9876543210	Generates 6-digit random OTP + Twilio SID	HTTP 200; SMS dispatched with unique SID	PASS
Payment Order Creation	Injects amount ₹800	Computes amount in paise (80000)	HTTP 200; Generates order order_hp_...	PASS
Payment Signature Verify	Injects Order & Payment IDs	Verifies signature integrity	HTTP 200; Returns digital receipt with PAID status	PASS

4. Conclusion & Verification

Experiment 08 demonstrated professional third-party API integration. By orchestrating Nodemailer for clinical prescriptions, simulating Twilio telecom protocols for OTP two-factor verification, and structuring Razorpay/Stripe transaction pipelines, the application verified end-to-end cloud microservice communication.